A Deep Reinforcement Learning Approach for Joint Task Offloading, Caching, and User Association in Mobile Edge Computing

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Introduction & Motivation

- Modern mobile apps (e.g., AR/VR, 3D games, recognition) are computation-heavy, while devices remain resource constrained.
- Mobile Edge Computing (MEC) pushes computation from devices to nearby edge nodes (APs / SBSs) to reduce latency.
- However, simultaneous offloading from many users causes:
  - Highly dynamic edge loads,
  - Congestion and deadline violations,
  - Inefficient use of limited radio and compute resources.
- We focus on **jointly** deciding:
  - Whether to offload a task or process locally,
  - Which AP(s) should serve each user,
  - Which files to cache / evict at each AP.

Problem Statement

- Edge nodes have limited storage, bandwidth, and compute.
- Pure task-offloading policies that ignore caching and AP load lead to:
  - Long transmission + queuing delay,
  - High task drop ratios when queues overflow.
- Existing DRL and game-theoretic methods often:
  - Assume single AP association per user,
  - Require global knowledge of the entire network state,
  - Do not scale well in ultra-dense MEC deployments.
- **Goal:** Learn a *decentralized* policy that coordinates caching, user–AP association, and task offloading to minimize end-to-end delay.

Key Contributions

- A **Deep Q-Network (DQN)** framework that couples:
  - User–AP association,
  - Cache placement / replacement,
  - Task offloading decisions.
- Allows each user to associate with **multiple APs** to fetch different files.
- Incorporates channel state information (CSI), interference, and cache status into the DRL state.
- Uses **Zipf** for initial caching and **Bernoulli** association to model realistic content requests.
- Demonstrates improved convergence, reduced delay, and lower task drop ratio compared with a state-of-the-art DRL baseline.

Related Work

- DRL-based offloading:
  - Deep RL for local/offload decisions and edge scheduling.
  - DQN for offloading with large state spaces and wireless charging.
- Caching replacement:
  - Greedy caching under matroid constraints for delay minimization.
  - DQN-based caching focusing mainly on delay but not multi-AP association.
- Game-theoretic models:
  - Weakly acyclic games and Two-Dimensional Nash Equilibria (2D-NE) for caching and MU–BS association.
- **Gap:** Existing methods typically treat offloading, caching, and association separately and often rely on centralized knowledge.

References

[1] F. Guo et al., "Joint optimization of caching and association in energy-harvesting-powered small-cell networks," *IEEE Trans. Veh. Tech.*, 2018.

[2] M. Tang and V. W. Wong, "Deep reinforcement learning for task offloading in mobile edge computing systems," *IEEE Trans. Mobile Comput.*, 2022.

[3] X. Chen et al., "Performance optimization in mobile-edge computing via deep reinforcement learning," arXiv:1804.00514, 2018.

[4] J. Wang et al., "Smart resource allocation for mobile edge computing: A deep reinforcement learning approach," *IEEE TETC*, 2021.

(Full bibliography available in the paper.)

System Model

Network:

- B APs (edge nodes), set $\mathcal{B} = \{1, \dots, B\}$.
- K mobile users, set $\mathcal{K} = \{1, \dots, K\}$.
- Content library with F files; each AP caches a subset of $J \ll F$ files.

Connectivity:

- Binary user–AP connectivity matrix $P(t) \in \{0,1\}^{K \times B}$: $P_{bk}(t) = 1$ if user k is in coverage of AP b.
- Binary caching matrix $Z(t) \in \{0,1\}^{F \times B}$: $Z_{fb}(t) = 1$ if file f is cached at AP b.

Requests:

- Each user requests R files, modeled by request vectors $u_{bk} \in \{0,1\}^F$, with $\|u_{bk}\|_1 = R$.
- Request popularity follows a Zipf distribution.

Wireless Model & Delay

Downlink rate (single AP):

$$R_{bk}(t) = W_{bk} \log_2 \left(1 + \frac{P_{bk}|h_{bk}(t)|^2}{N_0 + I_k}\right)$$

- W_{bk}: bandwidth share for user k at AP b.
- P_{bk}: transmit power; $h_{bk}(t)$: channel gain;
- N_0: noise PSD; I_k: interference from other APs.

Delay components:

- If file is cached at AP: delay $\propto \dfrac{|s_f|}{R_{bk}(t)}$.
- If file is not cached: additional backhaul / fetching penalty D_{cf}.

Cost (per AP b):

$$c_{3,t,b} = \sum_{f,k} \sigma_{bk}(t)\alpha_{bf}(t)u_{bfk}\frac{|s_f|}{R_{bk}(t)} + \sum_{f,k} \sigma_{bk}(t)(1-\alpha_{bf}(t))$$

where $\sigma_{bk}(t)$ is the association indicator and $\alpha_{bf}(t)$ is the caching indicator.

DRL Formulation (Per AP b)

State:

$$x_b(t) = (\alpha_b(t), \sigma_b(t), G_b(t))$$

- $\alpha_b(t)$: cache state (F-dimensional binary vector),
- $\sigma_b(t)$: user–AP association (K-dimensional binary vector),
- $G_b(t)$: CSI / channel quality indicators.

Action:

$$y_b(t) = (\sigma_b(t), \alpha_b(t))$$

- Joint decision: which users to serve and which files to cache/evict.

Reward:

$$r_b(t) = -c_{3,t,b}$$

- Negative total delay encourages low-latency policies.

Objective:

- Learn Q-network parameters θ to minimize discounted long-term cost:

$$\min_{\pi} \ \mathbb{E}\left[\sum_t \gamma^t c_{3,t,b}\right]$$

Enhanced DQN Algorithm

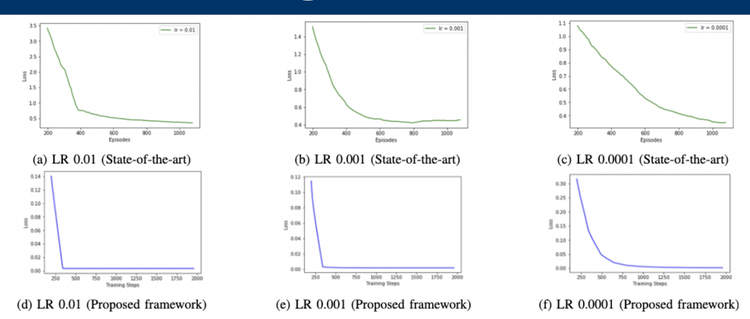
Key Features

- Replay buffer with capacity limit and mini-batch sampling.
- Separate **primary** and **target** Q-networks with periodic soft/hard updates.
- ϵ-greedy exploration with initial ϵ, minimum $\epsilon_{\min}$, and decay factor.
- Actions jointly encode user–AP associations and cache replacement.

High-Level Loop

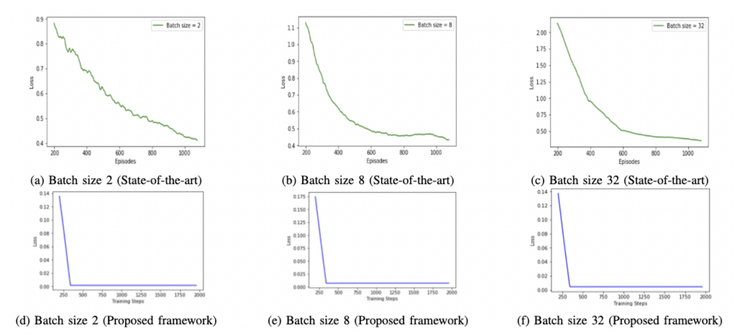
1. Observe current state $x_b(t)$ (cache, associations, CSI).
2. Choose action $y_b(t)$ via ϵ-greedy over Q-values.
3. Apply action: update cache and association; compute delay-based cost.
4. Store $(x_b(t), y_b(t), r_b(t), x_b(t+1))$ in replay buffer.
5. Sample mini-batch, compute targets, and update θ by gradient descent.
6. Periodically update target network parameters and decay ϵ.

Results: Convergence Behavior



(a) LR 0.01 (State-of-the-art) (b) LR 0.001 (State-of-the-art) (c) LR 0.0001 (State-of-the-art)
(d) LR 0.01 (Proposed framework) (e) LR 0.001 (Proposed framework) (f) LR 0.0001 (Proposed framework)

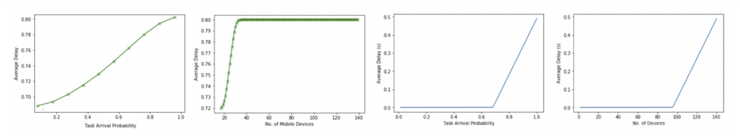
Convergence under different learning rates.



(a) Batch size 2 (State-of-the-art) (b) Batch size 8 (State-of-the-art) (c) Batch size 32 (State-of-the-art)
(d) Batch size 2 (Proposed framework) (e) Batch size 8 (Proposed framework) (f) Batch size 32 (Proposed framework)

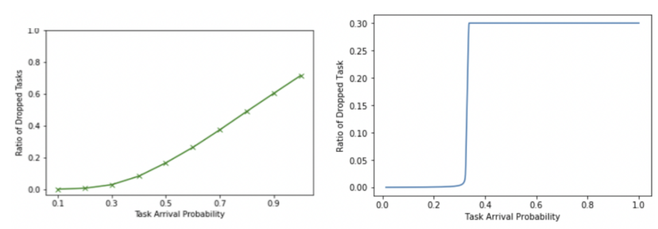
Convergence under different batch sizes.

Results: Delay & Dropped Tasks



(a) Average delay vs task arrival probability (state-of-the-art) (b) Average delay vs no. of devices (state-of-the-art) (c) Average delay vs task arrival probability (proposed framework) (d) Average delay vs no. of devices (proposed framework)

Average delay vs. TAP and number of devices.



(a) State-of-the-art Solution (b) Proposed Solution

Dropped task ratio vs. TAP.